# 02. Roman Numerals & Negative Numbers

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

## Worked Examples

**Example 1 — Writing a Roman numeral:**

Write 34 as a Roman numeral.

**Answer:** Split into tens and units: 34 = 30 + 4. 30 = XXX. 4 = IV (one less than five). So 34 = **XXXIV**.

**Example 2 — Reading a Roman numeral:**

What number does XLVII represent?

**Answer:** Split it up: XL = 40 (X before L means subtract: 50 − 10 = 40). VII = 7. Add them: 40 + 7 = **47**.

**Example 3 — Falling through zero, step by step:**

The temperature is 4°C. It falls by 10°C. What is the new temperature?

**Answer:** Count down ten steps from 4: 4, 3, 2, 1, 0, -1, -2, -3, -4, -5, -6. Or simply: 4 − 10 = **-6°C**.

**Example 4 — Ordering negative numbers:**

Order from coldest to warmest: -3°C, 5°C, -8°C, 0°C.

**Answer:** The more negative a number is, the colder it is — so -8 is colder than -3. Working from coldest: **-8°C, -3°C, 0°C, 5°C**.

**Example 5 — Select-all, with full reasoning:**

Which of these are TRUE? Select ALL that apply. A) LXI represents 61 B) The Roman numeral for 49 is IL C) A fall from 2°C to -5°C is a drop of 7 degrees D) -10°C is colder than -6°C

**Answer:** A: LX = 60, I = 1, so LXI = 61 — TRUE. B: 49 is written XLIX (40 + 9); 'IL' is not a valid Roman numeral (you can only subtract I from V or X, never from L) — FALSE. C: 2 − (-5) = 7 — TRUE. D: -10 is further below zero than -6, so it is colder — TRUE. Answer: **A, C and D**.

## Foundation

*15 questions*

**1.** Write 6 as a Roman numeral.

**2.** What number does XIV represent?

**3.** Write 17 as a Roman numeral.

**4.** What number does XI represent?

**5.** Which Roman numeral represents 9? A) VIIII B) IX C) XI D) IV

**6.** Write 13 as a Roman numeral.

**7.** What number does VIII represent?

**8.** Write 19 as a Roman numeral.

**9.** The temperature is 5°C. It falls by 3°C. What is the new temperature?

**10.** The temperature is -2°C. It rises by 6°C. What is the new temperature?

**11.** Which is colder, -4°C or -1°C?

**12.** The temperature is 1°C. It falls by 4°C. What is the new temperature?

**13.** Order from coldest to warmest: 2°C, -3°C, 0°C.

**14.** The temperature starts at 0°C and falls by 7°C. What is the new temperature?

**15.** The temperature is -5°C. It rises by 2°C. What is the new temperature?

## Standard

*20 questions*

**1.** Write 28 as a Roman numeral.

**2.** What number does XXXVI represent?

**3.** Which Roman numeral represents 40? A) XXXX B) XL C) LX D) XC

**4.** Write 62 as a Roman numeral.

**5.** What number does XLIV represent?

**6.** Select ALL of the Roman numerals below that represent numbers greater than 50: A) XLVIII B) LXI C) XLIX D) LVII E) XXXIX

**7.** Write 86 as a Roman numeral.

**8.** What number does LXXIX represent?

**9.** Which Roman numeral represents 94? A) XCIV B) IC C) LXXXXIV D) XCVI

**10.** Write 58 as a Roman numeral.

**11.** The temperature is 6°C. It falls by 11°C. What is the new temperature?

**12.** Order from coldest to warmest: -6°C, 4°C, -1°C, -9°C.

**13.** What is the difference between 5°C and -4°C?

**14.** The temperature is -7°C. It rises by 12°C. What is the new temperature?

**15.** Which is warmest? A) -3°C B) -8°C C) 1°C D) -1°C

**16.** The temperature starts at -3°C, falls by 5°C, then rises by 4°C. What is the final temperature?

**17.** A submarine is at a depth of -25 m. It rises 9 m. What is its new depth?

**18.** Select ALL of the following that are TRUE: A) A rise from -6°C to 2°C is a rise of 8 degrees B) -12°C is warmer than -9°C C) The difference between 3°C and -3°C is 6 degrees D) 0°C is colder than -1°C

**19.** Write the Roman numeral for seventy-seven.

**20.** A lift starts on floor 2. It goes down 6 floors. What floor is it on now?

## Hard

*5 questions*

**1.** A theatre seat is labelled with the Roman numeral LXXXVIII. What seat number is this?

**2.** Order from coldest to warmest: -11°C, 3°C, -4°C, -1°C, 7°C.

**3.** A lift is on floor -3 (three floors below ground). It rises 8 floors, then falls 2 floors. What floor is it on now?

**4.** Which Roman numeral is written correctly for 99? A) IC B) XCIX C) LXXXXIX D) VC

**5.** The temperature at the top of a mountain is -14°C. Halfway up, it is 11°C warmer than the top. At the base, it is 9°C warmer than halfway up. What is the temperature at the base?

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